Aurora vs RDS — When to Choose What

Comparison Table

Feature	RDS MySQL	Aurora MySQL
Max Storage	Up to 64 TiB	Up to 128 TiB (auto-growing)
Storage Scaling	Manual or storage auto-scaling on provisioned EBS storage	Automatically grows in 10 GB increments up to 128 TiB
Read Replicas	Up to 15 read replicas (asynchronous replication)	Up to 15 Aurora Replicas sharing the same cluster storage
Failover Time	Typically 60–120 seconds for Multi-AZ deployments	Typically less than 30 seconds when a replica is available
Multi-AZ Cost	Requires a standby instance, increasing cost significantly	Multi-AZ storage replication is built in; additional cost mainly comes from extra reader instances
Global Database	Cross-Region Read Replicas available; promotion is manual	Native Aurora Global Database with cross-region replication and typically sub-second lag
Serverless Option	No native MySQL serverless deployment	Aurora Serverless v2 available with fine-grained automatic scaling
Relative Cost	Lower cost, suitable for budget-conscious workloads	Higher cost, optimized for performance and availability
MySQL Compatibility	Full MySQL compatibility	MySQL-compatible (Aurora MySQL)
Best For	Traditional applications, smaller workloads, cost-sensitive projects	High-performance production systems, large-scale applications, global deployments, and workloads requiring fast failover

When to Choose RDS MySQL

Choose RDS MySQL when:

- Cost is a major consideration.
- You need a managed MySQL database without advanced Aurora features.
- Your workload is small to medium-sized.
- Standard Multi-AZ and read replica capabilities are sufficient.

When to Choose Aurora MySQL

Choose Aurora MySQL when:

- High availability and rapid failover are critical.
- You need better read scalability.
- Your application serves users globally.
- You want automatic storage growth and serverless scaling.
- Performance requirements justify the additional cost.

Key Takeaway

For most small and medium applications, RDS MySQL provides excellent value and simplicity. For mission-critical production workloads that require higher performance, faster failover, global replication, and elastic scaling, Aurora MySQL is usually the better choice despite the higher cost.